Higher National Diploma in Quantity Surveying

02nd Year, First Semester Examination - 2017

QS 21013 – Measurement II

Instructions for Candidates:
Answer four question only

No. of questions : 05
No. of pages: 06
Time: 03 hours

Instruction to Candidates:

- Answer question 01 (compulsory) and any Three (03) questions from 2-5
- Drawings are enclosed separately and specification details are given
- Taking off quantities should be in standard TDS sheets
- Where any information is missing, the specification is inadequate, or exact nature of construction is unclear, you are required to make sensible assumptions in relation to the measurement. Any assumptions made shall be entered in to query sheet, for the attention of Architect/Engineer.
- Clean unmarked copy of SLS 573 is permitted to answer the exam paper

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1.

I.

a. Briefly explain the term **Provisional sum** (03 marks)

b. What are the types of information that have to be provided with provisional sum? (04 marks)

c. Give three (03) examples for the provisional sum for works (03 marks)

II.

a. What is the meaning of **preliminaries?** (02 marks)

b. Give three examples for preliminaries (03 marks)

III. What is **the purpose of a BOQ?** (05 marks)

IV. Briefly explain the role of Quantity Surveyor after awarding the contract (Post contact stage) (05 Marks)

(Total 25 marks)

2.

Refer the drawing named as DWG 01 (Column and footing details) for the question 2.
Taking off quantities of following sub structures (There are 20 column footings)

a. Excavation for the column footings (05 marks)
b. Screed concrete [1:3:6 (25)] (05 marks)
c. Concrete and formwork of the column bases (07 marks)
d. Concrete and formwork of the column up to finish floor level (FFL) (08 marks)

(Total 25 marks)

3.
Refer the drawing named as DWG 02 for the question 3.

I. Taking off quantities of following supper structures
   a. Concrete and formwork of columns up to roof slab (06 marks)
   b. Concrete and formwork of beams at the roof slab level (06marks)
   c. Concrete and formwork of roof slab (06 marks)

II. Prepare a BOQ for the above details giving descriptions as per the standards (07marks)

(Total 25 marks)

4.
Refer the drawing named as DWG 02 for the question 4.

Prepare the Bar Schedule for the Reinforcements

a) Column base
b) Column
c) Beam
d) Slab (25 marks)

Diameter	Weight (Kg/m)
32	6.31
28	4.83
25	3.85
22	2.98
20	2.47
16	1.58
12	0.89
10	0.23

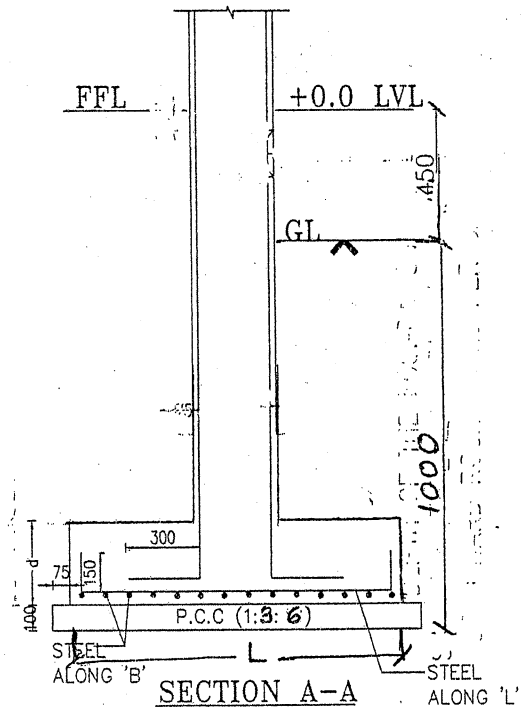
5.

Refer the drawing named as DWG 03 for the question 5.

Taking off quantities for the following steel structures

a.	Wall plates	(4 Marks)
b.	Trusses	(4 Marks)
c.	Ridge	(4 Marks)
d.	Roof covering	(4 Marks)
e.	Ridge tiling	(3 Marks)
f.	Prepare a draft bill for above items	(6 Marks)
		(Total 25 marks)

COLOUMN AND FOOTING DETAILS



SECTION A-A

Footing Type	Length of footing (L)	Width of footing (B)	d	Column size	Footing Nos.	Footing Reinforcement	
						AST1	AST2
F1	1400	1400	230	450x230	10	Y10-150 c/c	Y10-150 c/c
F2	1600	1600	230	450x230	6	Y10-150 c/c	Y10-150 c/c
F3	1900	1900	230	450x230	4	Y10-150 c/c	Y10-150 c/c

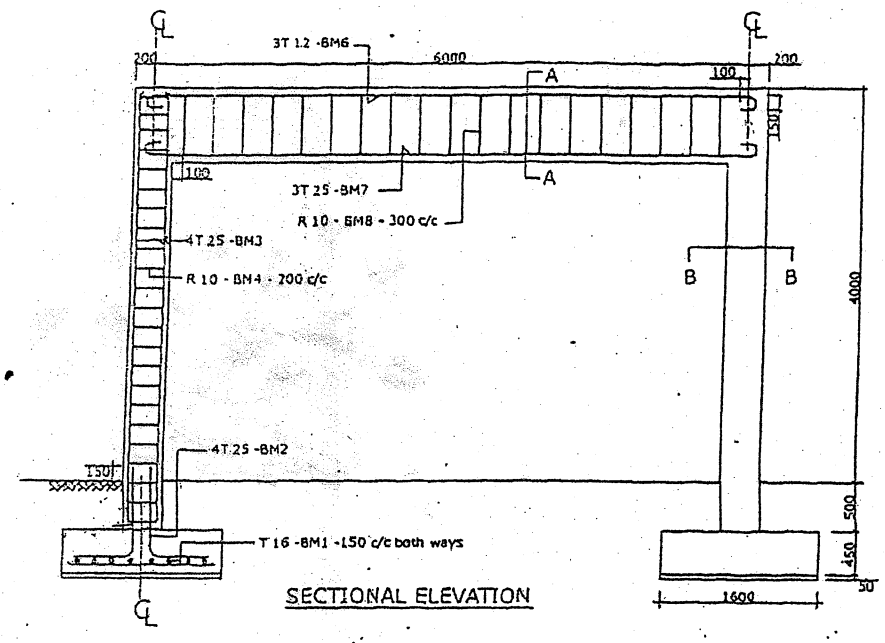
NOTES:

1. ALL DIMENSIONS ARE IN MM, AND ALL LEVELS ARE IN METERS UNLESS OTHERWISE NOTED.
2. DON'T SCALE THE DRAWING. FOLLOW THE WRITTEN DIMENSIONS
3. GRADE OF CONCRETE:–
   FOR FOOTINGS : M25
   COLUMNS : M25
   SLABS & BEAMS : M25
4. USE Fe 500 GRADE OF STEEL CONFIRMING TO IS:1786
5. CLEAR COVER TO STEEL:
   FOR FOOTINGS : 50mm
   FOR COLUMNS : 40mm
   FOR BEAMS : 30mm
   FOR SLABS : 25mm
6. THE SBC OF SOIL SHALL NOT BE LESS THAN 25 T/M² FOUNDING LEVEL
7. LOADING:–

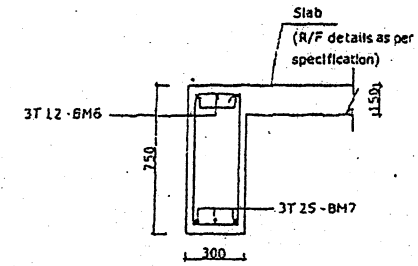
TYPE OF FLOOR	SUPER IMPOSED LOAD (KN/M)²	LIVE LOAD (KN/M)²	FLOOR FINIS (KN/M)²
FIRST FLOOR	1.0	3	1.5

DWG 01

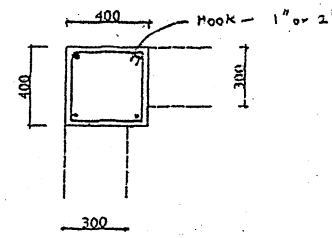
DWG 02



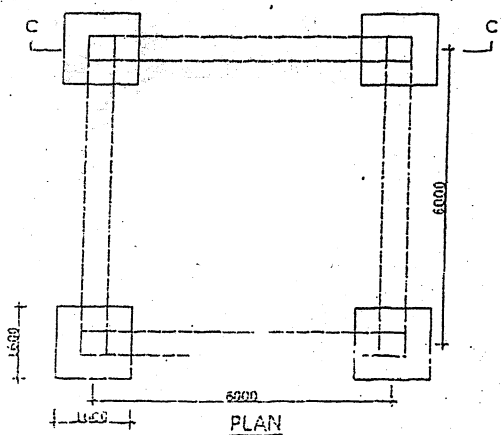
SECTIONAL ELEVATION



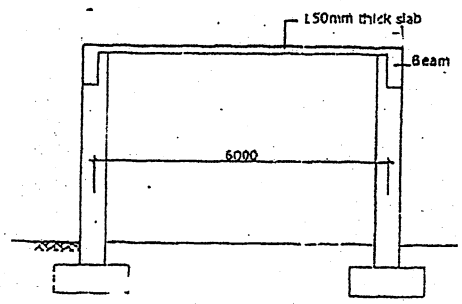
SECTION A-A



SECTION B-B



PLAN



SECTIONA

SPECIFICATION NOTES

Reinforcement -
All R/F to be accordance with BS4449
and characteristic strength of 250N/mm2
for mild steel and 410N/mm2 for tor steel

Column base	50mm clear cover
Column	30mm clear cover
Beam	30mm clear cover
Slab	12mm Ø bars @ 110mm c/c both directions 30mm clear cover

Note:
All dimensions are not to the scale

PROJECT NAME: REINFORCEMENT - 01
DRG. NO. EX-RF-01

DWG 02

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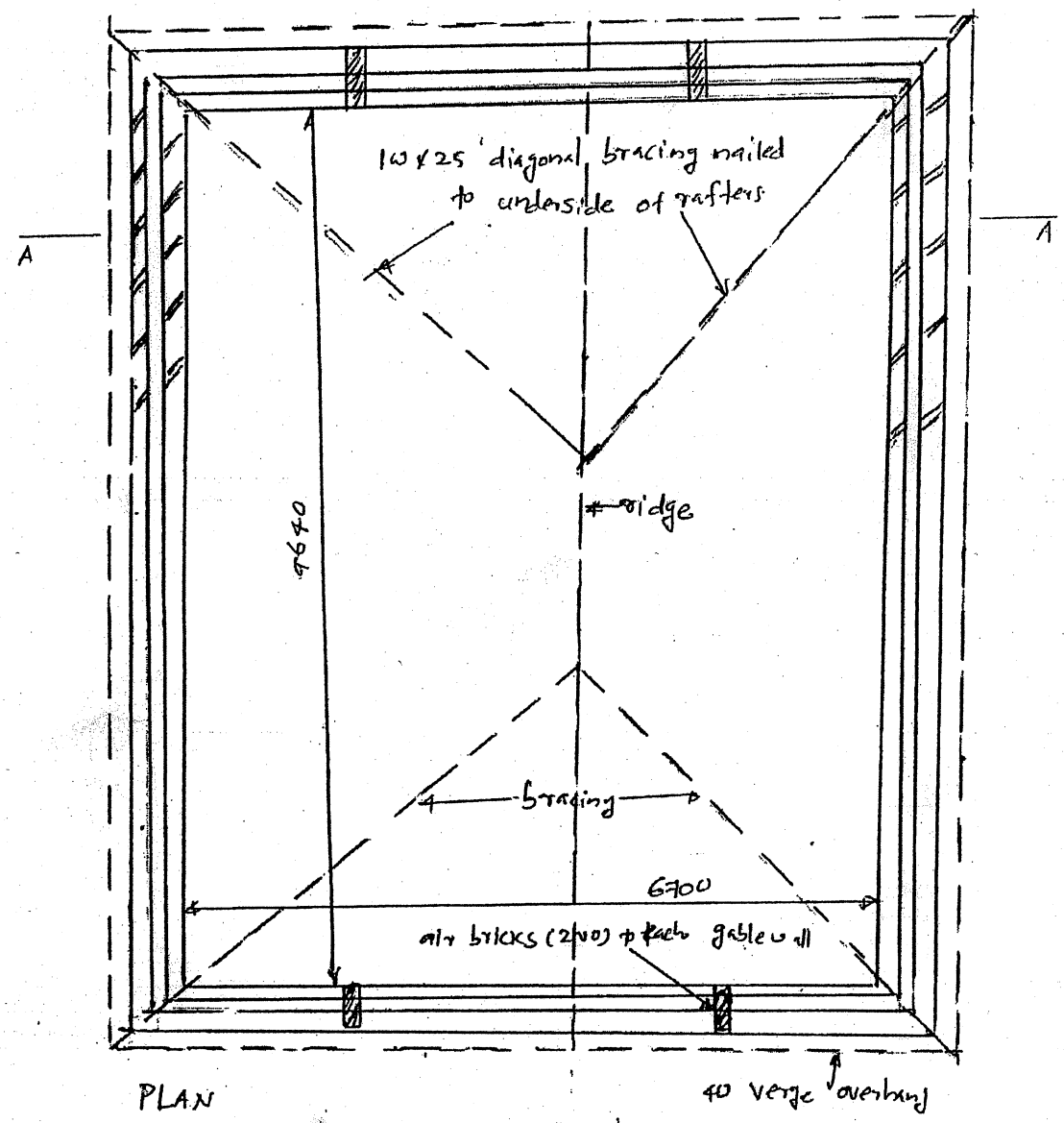
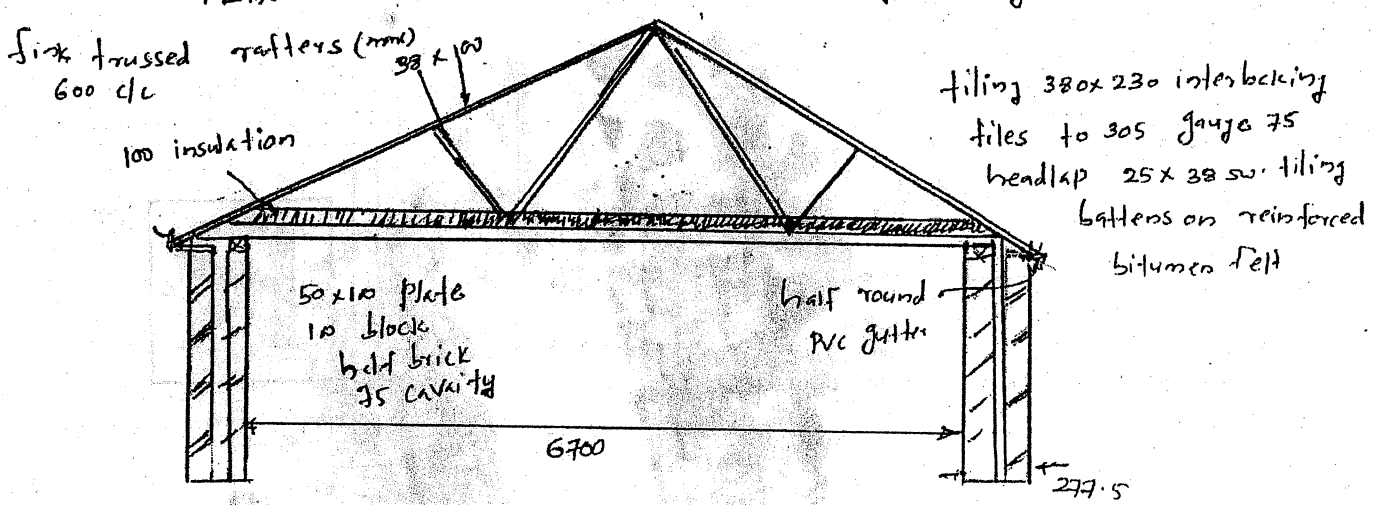
DWG 03



PLAN



SECTION A-A